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Payload Bay Venting in Flight (rev. 2)

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Introduction

Knowledge of the internal pressure in a sounding rocket is important for several reasons. First, if the internal pressure differs from that outside in the atmosphere, the payload shell will be in tension. This can, in extreme cases, cause structural failure[1]. Another issue is measurement of atmospheric pressure inside a payload cavity. In both cases one would like to have the internal pressure match as closely as possible the external atmospheric pressure. We assume that a system of vents between the payload cavity and the atmosphere has been provided with all vents having the same external pressure.

This memo documents an EXCEL code called BLOWDOWN2.xls.

Assumptions

While the analysis that follows is simplistic, it captures the major features of the relevant phenomenology. First, note that the vent throat Mach number can be subsonic or sonic (choked) depending on the vent pressure drop. Second, assume that the pressure coefficient at the vent exit is a known function of flight Mach number. Suppose we wish to locate the vent exit so that its exit pressure coefficient vanishes. This generally implies that the vents be placed well aft of the rocket nose, but not too close to its fins. Since air can flow into the payload as well as out, the same orifice coefficients are used for both cases. Third, assume that the vent holes are cylindrical. This implies their exit and throat areas are the same. Fourth, assume the flow through a vent is adiabatic and isentropic. Fifth, absent any additional sources of gas in the payload cavity, we assume the air trapped in the payload cavity remains stagnant; i.e., has negligible velocity as it expands. Sixth, assume the air trapped inside a payload cavity expands isentropically as the rocket ascends. Finally, throughout, we assume that the trapped air behaves as perfect gas.

Notation

V = Velocity in the vent,
U = Free stream velocity,
h = Altitude (MSL),
p = Pressure,
ρ = Mass density of air,
T = Temperature of air inside the payload cavity,
R = Gas constant for air = 1716.4827 $ft^2/sec^2\ ^oR$,
γ = Ratio of specific heats of air = 1.4,

[1] A dramatic example was the first test launch of the Space General Astrobee 1500 in which test instrumentation indicated the failure happened early in first stage (a Sergeant solid rocket) burn. Definitive evidence came from ballistic camera imagery that showed a large black area where part of the payload shroud had been.

M = Mach number,
a = Sound speed,
A = Area of the vent throat,
Vol = Volume of the payload cavity,
C_p = Pressure coefficient,
$C_p(0)$ = Incompressible pressure coefficient,
C_N = Nozzle orifice coefficient,
C_{SON} = Nozzle orifice coefficient for sonic (choked) flow,
C_{INC} = Nozzle orifice coefficient for incompressible flow,
C_{SUB} = Nozzle orifice coefficient for subsonic flow,
t = Time,
$( )_C$ = Value of ( ) inside the cavity,
$( )_T$ = Value of ( ) in the vent throat,
$( )_a$ = Ambient atmospheric value of ( ),
$( )_\infty$ = Free stram value of ( ),
$( )_X$ = Value of ( ) at the vent exit, and
$( )_0$ = Value of ( ) in the cavity at liftoff.

Introduction

For the venting under discussion, consider the steady, isentropic flow of a perfect gas. The mass flow rate through a vent is given by

$$\frac{dm}{dt} = -\rho_T V_T A\, C_N \tag{1a}$$

Several comments on eq. (1a) are in order. First, it doesn't matter whether the vent area is for a single vent hole or is the area sum for multiple holes as long as the vent exit pressures are the same for all. Good engineering practice suggests that a minimum of three vents, equally spaced in azimuth, be provided to mitigate first order angle of attack biases. Second, the factor C_N is an empirical correction to the theoretical mass flow rate.

Next, the mass flow rate out the vent must be matched by the rate of mass loss inside the cavity:

$$\frac{dm}{dt} = Vol\,\frac{d\rho}{dt} \tag{1b}$$

The essence of this problem is matching the two parts of eq.(1), and numerically integrating the result to find the cavity pressure history.

Nozzle Orifice Coefficient

First, consider incompressible vent flow. For a sharp edged hole, C_N can be as low as ~½. But, if the upstream corner contour is smoothly radiused, C_N can approach 0.8. For choked flow, C_N approaches 0.96-0.98.

With these assumptions we can use the results in ref. (1), eq. (44), to calculate the vent pressure ratio for sonic (choked) flow in the throat:

$$\frac{p_X}{p_C} \le \left(\frac{2}{\gamma+1}\right)^{\left(\frac{\gamma}{\gamma-1}\right)} \tag{2}$$

As long as eq.(2) is satisfied the vent throat will have choked flow $(M_T = 1)$, and its orifice coefficient C_N will remain constant and $= C_{SON}$. For subsonic throat flow the orifice coefficient will vary smoothly with throat Mach number from its incompressible value to its value for choked flow. The throat Mach number is found from ref.(1), eq. (44), to be

$$M_T = \sqrt{\left(\frac{2}{\gamma - 1}\right)\left[\left(\frac{p_C}{p_X}\right)^{(\gamma-1)/\gamma} - 1\right]} \tag{3}$$

The subsonic orifice coefficient is based on a hodograph solution by Busemann, see ref. (2), for a two dimensional orifice. The Busemann interpolator is

$$C_{SUB} = \frac{C_{SON}}{\left(1 + \beta\left(\left(\frac{C_{SON}}{C_{INC}}\right) - 1\right)\right.} \quad \text{, where} \tag{4}$$

$$\beta = \sqrt{1 - M_T^2}\,.$$

Thus, one inputs C_{INC} and C_{SON}, computes the throat Mach number, and finds the nozzle orifice coefficient. See ref. (4) for a survey of incompressible orifice coefficients.

External Pressure

The vent external pressure is found from the external rocket aerodynamics, and defined in terms of a pressure coefficient C_{pX}:

$$p_X = p_a + \frac{1}{2}\rho_a U^2 C_{pX} \tag{5}$$

It's necessary to start with a numerical solution for the trajectory variables, altitude h and velocity U. When these are known, the atmospheric pressure p_a and density ρ_a, both functions of altitude alone, can readily be found. The exit pressure coefficient C_{pX} is customarily considered to be a function of free stream Mach number. It includes the

effects of both general body shape and any devices near the vent exit intended to influence venting behavior. For subsonic free stream Mach numbers, one first finds the incompressible value of C_{pX} using the methods of ref. (6). Then, based on ref. (5), use the Karman-Tsien correction for subsonic pressures:

$$C_{pX}(M_\infty) = \frac{C_{px}(0)}{\sqrt{1 - M_\infty^2} + \frac{C_{px}(0)}{2}(1 - \sqrt{1 - M_\infty^2})} \tag{6}$$

The Karman-Tsien correction is a good approximation so long as the local flow over the rocket body surface is subsonic everywhere. Local supersonic flow leads to shock waves and a flow pattern unlike that for incompressible flow. To ensure this does not happen, find the minimum incompressible pressure coefficient from MUNKSHIP2 and plug it into CRITICAL MACHNUMBER. Keep in mind that the vent location under analysis may not be that for the critical Mach number.
When the free stream Mach number is supersonic, a different approach is needed. Estimate the surface pressure coefficient C_{pX} using ref. (7) as a guide. Engineers often develop "eyeball" estimates for the transonic surface pressure coefficient by laying their favorite french curve between the subsonic and supersonic data points. The current version of BLOWDOWN2 uses eq.(6) for Mach numbers less than the critical Mach number, M_{cr} ,the free stream Mach number for which the local Mach first reaches unity. For Mach numbers greater than critical, BLOWDOWN2 uses linear interpolation to estimate an exit pressure coefficient.

Nozzle Mass Flow Rate

Equation (1a) can also be written as

$$\frac{dm}{dt} = -\rho_T a_T M_T A C_N$$

Again, from ref.(1), eq's. (45) and (46)for isentropic nozzle flow, give

$$\rho_T a_T = \rho_C a_C \left[1 + \left(\frac{\gamma - 1}{2}\right) M_T^2\right]^{\frac{-(\gamma+1)}{2(\gamma-1)}}$$

Since the air trapped in a payload bay expands isentropically from its initial state,

$$\rho_T a_T = p_0 \sqrt{\frac{\gamma}{RT_0}} \left(\frac{p_C}{p_0}\right)^{\frac{\gamma+1}{2\gamma}} \left[1 + \left(\frac{\gamma - 1}{2}\right) M_T^2\right]^{\frac{-(\gamma+1)}{2(\gamma-1)}}$$

(7)

Substituting eq's.(3) and (7) into the modified eq.(1) allows the nozzle mass flow rate to be found in terms of the vent throat Mach number and payload cavity pressure.

Payload Bay Expansion

The mass of trapped air inside the payload bay is just $\rho_C Vol$. The relationship we need between pressure p and density ρ is just that for an isentropic process:

$$\frac{\rho_C}{\rho_0} = \left(\frac{p_C}{p_0}\right)^{\frac{1}{\gamma}} \tag{8}$$

Here, the subscript **o** refers to conditions inside the payload bay at liftoff. Then,

$$\frac{d\rho_C}{dt} = \left(\frac{\rho_0}{\gamma}\right)\left(\frac{p_C}{p_0}\right)^{\frac{1}{\gamma}}\left(\frac{1}{p_C}\right)\frac{dp_C}{dt}$$

$$= \frac{\rho_0}{\gamma p_0^{\frac{1}{\gamma}}} p_C^{-(\gamma-1)/\gamma} \frac{dp_C}{dt}$$

Nozzle-Cavity Matching and Integration

Clearly the rate of change of mass of trapped air must equal that lost through the vent hole(s) as captured in eq. (1b). Matching the two mass rates of change gives:

$$\frac{dp_C}{dt} = -\gamma\sqrt{\gamma R T_0}\left(\frac{A}{Vol}\right) M_T C_N p_0^{\frac{-(\gamma-1)}{2\gamma}} p_C^{\frac{(3\gamma-1)}{2\gamma}} \left[1+\left(\frac{\gamma-1}{2}\right)M_T^2\right]^{\frac{-(\gamma+1)}{2(\gamma-1)}} \tag{9}$$

Equation (9) is a nonlinear first order differential equation for the cavity pressure. Note that the first part of the right hand side is a constant for every case, and need be evaluated only once at the start of each case.

In the event that $\frac{p_X}{p_C} > 1$, eq.(9) remains the same except that the leading "–" sign is replaced with a "+" sign.

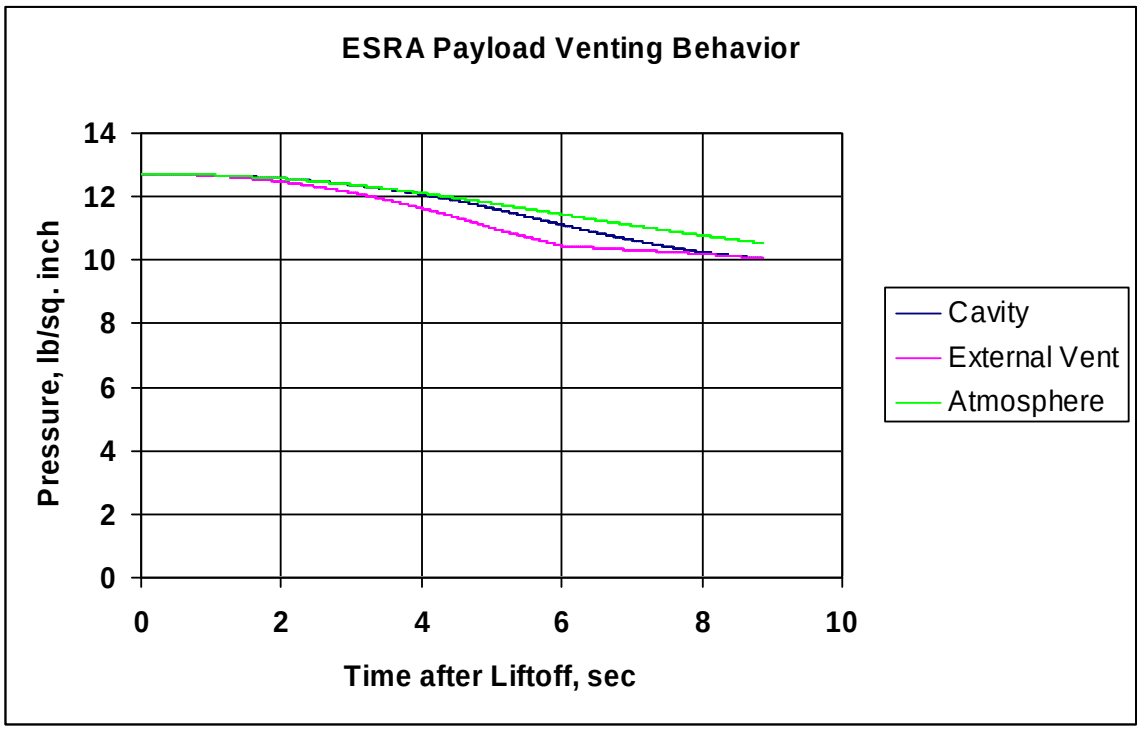
BLOWDOWN2.xls

BLOWDOWN2.xls uses a fourth order Runge-Kutta method to perform the needed numerical integration for p_C as a function of time. See ref. (3) for a description of this algorithm. All cells colored with a robin's egg blue require user inputs. The magenta cells are natural constants, and the white cells are generated by BLOWDOWN2.

The trajectory altitude and velocity are input as tabular functions of time after liftoff. The pressure coefficient at the vent exit is input as a tabular function of free stream Mach number. BLOWDOWN2 uses linear interpolation to establish the instantaneous values in the code. Trajectory data must be obtained from a trajectory simulation such as SKYAERO, and the vent external pressure coefficient must be found from an aerodynamic code.

The numerical example reported here is for an ESRA IREC sounding rocket designed to have an apogee of exactly 10,000 feet AGL. Its solid propellant motor burns out at ~6 seconds after liftoff.

The first graph shown below displays the history of atmospheric pressure, the pressure at the vent exit, and the payload cavity pressure. The atmospheric pressure decreases with the altitude, but the pressure at the vent exit decreases even more, especially near burnout. This is typically the case since most vents are located just behind the base of the nose cone where the subsonic pressure coefficient is usually negative. This case used an incompressible vent pressure coefficient adjusted using the Karman-Tsien method for compressibility. The critical Mach number for this location is about 0.85, and since the burnout Mach number is about 0.8, no shock effects are expected.

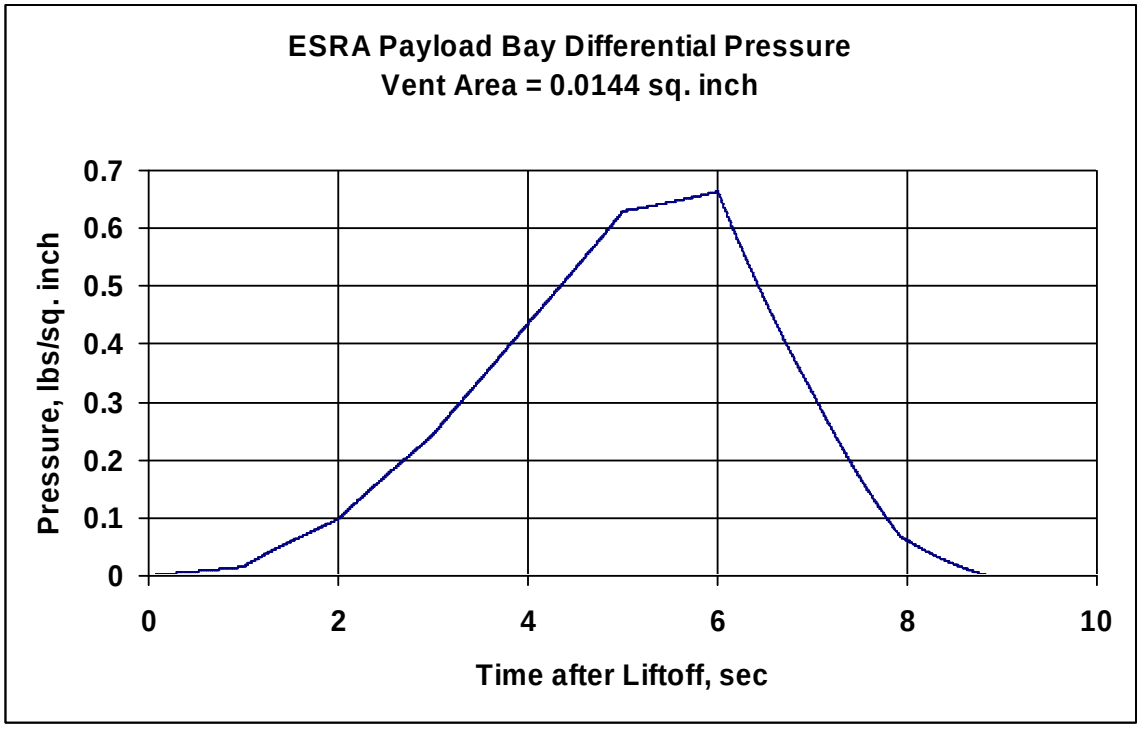


Next, explore the pressure difference between the payload cavity and the external pressure near the vent. This difference can be used to support a stress analysis of the payload shell, or to size the vent holes to satisfy a maximum Δp requirement.

Note the angular nature of this plot from 5 to about 7 seconds. This is an artifact of the granularity (1 second) of the SKYAERO trajectory data used. A more extensive trajectory table with finer granularity would give a smoother result.

The next study done was to assess the impact of various integration step sizes. It turns out that the maximum step size is a function of vent throat area. Large vents depressurize a payload bay very quickly. If the step size is too great, the Runge-Kutta integrator will overshoot the exit pressure resulting in flow from the outside back into the cavity,

something not reasonable. A step size of about 0.005 sec. will provide the results shown here.



In the example, the maximum pressure difference is about 0.6638 psi at 6.0 sec after liftoff (burnout).

References

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